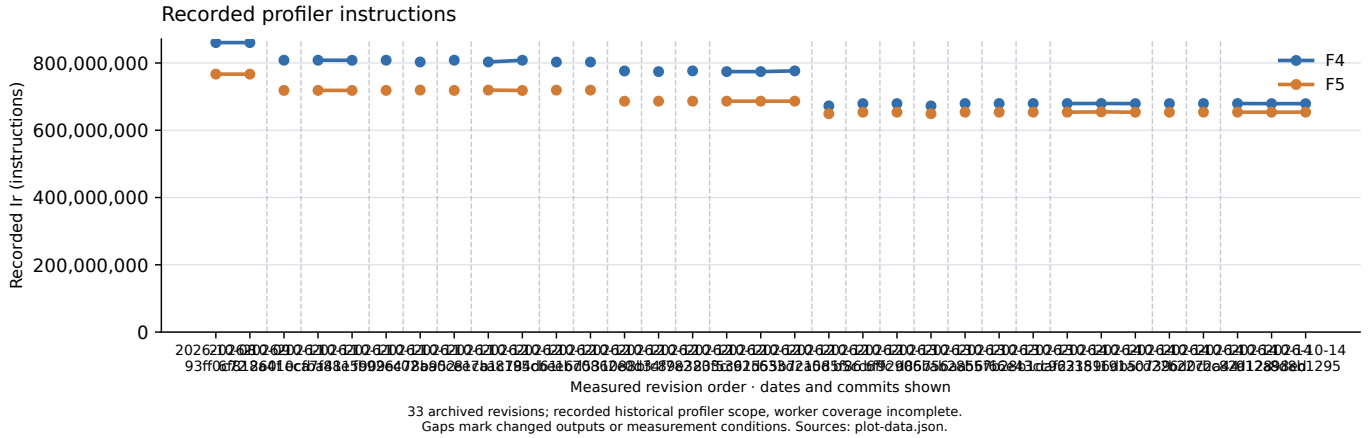
Boolean matrix step: F4 and F5

One shared K_1 / 2^23 m=2 system, 22 Boolean variables, maximum degree 4. Matrix construction, reduction and readback; F5 includes its criterion.





UTC date / revision	F4 Ir	F5 Ir	F4 ms	F5 ms
2026-10-08 / 93ff0c72	860,633,889	766,940,469	249.241	208.250
2026-10-09 / 6f81260f	860,555,567	766,776,087	333.279	264.687
2026-10-11 / 8a41eca7	808,263,705	718,576,235	97.380	94.658
2026-10-11 / 0cfba411	808,263,705	718,578,523	86.390	82.214
2026-10-11 / f88e5909	807,989,935	718,391,055	82.597	81.948
2026-10-11 / fb926c02	808,263,705	718,576,189	101.379	97.323
2026-10-11 / e478b952	803,017,657	719,516,063	107.585	102.927
2026-10-11 / aa0cee7b	808,281,869	718,492,761	58.723	60.673
2026-10-12 / 81ca1c78	803,015,877	719,509,003	100.762	98.775
2026-10-12 / a8194d61	808,072,539	718,282,030	101.827	96.904
2026-10-12 / 5cbe16d5	802,824,065	719,316,911	81.784	83.144
2026-10-12 / eb70862e	802,824,065	719,316,876	63.944	62.505
2026-10-12 / 31088bfc	776,242,363	686,315,894	102.929	99.489
2026-10-12 / 0d34f782	774,379,185	686,521,131	96.521	93.790
2026-10-12 / 89e3233c	776,490,367	686,543,371	81.628	80.581
2026-10-12 / 80f5391d	774,379,185	686,519,896	98.501	93.239
2026-10-12 / c62555b7	774,379,185	686,519,896	97.360	94.117
2026-10-12 / 633d2158	776,467,358	686,512,000	96.637	94.008
2026-10-12 / ca0d5f86	672,500,776	649,174,334	95.848	99.895
2026-10-12 / b5cc6ffc	679,242,389	653,622,849	49.965	52.882
2026-10-13 / bf92906b	679,242,389	653,622,849	90.163	90.201
2026-10-13 / d85756aa	672,486,486	649,160,075	95.497	95.480
2026-10-13 / ab28b57b	679,449,939	653,823,121	87.697	87.878
2026-10-13 / 56f66e43	679,449,939	653,828,331	90.448	94.715
2026-10-13 / 28b1daf2	679,449,939	653,828,331	56.215	57.961
2026-10-14 / cc962159	679,449,939	653,831,656	88.546	90.149
2026-10-14 / 33816db5	679,549,997	654,627,509	88.344	90.874
2026-10-14 / 191a0739	679,240,609	653,664,759	85.135	87.917
2026-10-14 / cd226d0c	679,240,609	653,664,759	92.059	98.073
2026-10-14 / b2272a44	679,449,939	653,873,401	78.007	85.621

UTC date / revision	F4 Ir	F5 Ir	F4 ms	F5 ms
2026-10-14 / bc82d128	679,449,939	653,873,401	86.126	89.102
2026-10-14 / 912a98ed	679,098,623	653,830,968	89.995	91.125
2026-10-14 / 8d8b1295	679,098,623	653,830,968	89.395	90.230

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

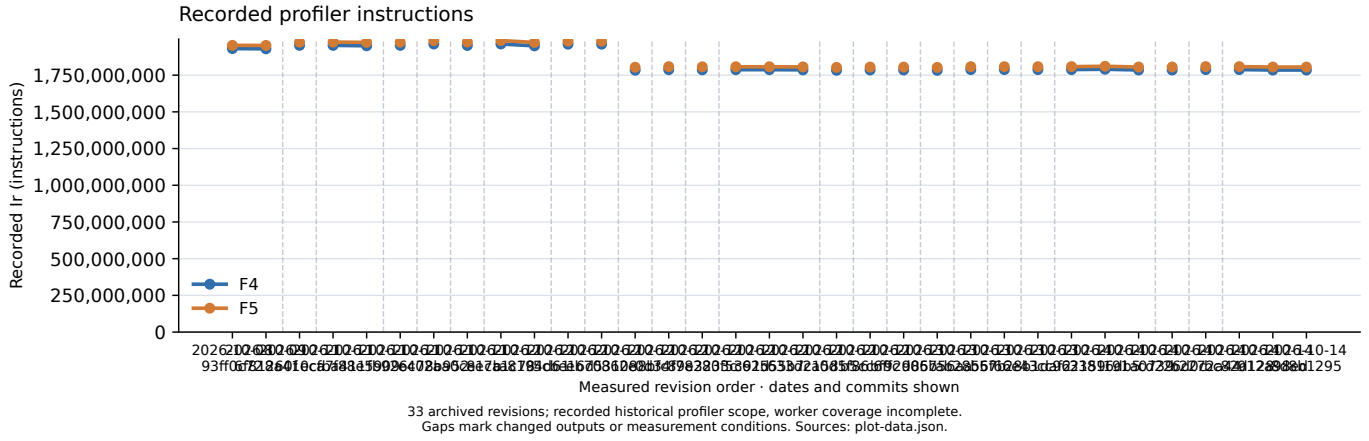
Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

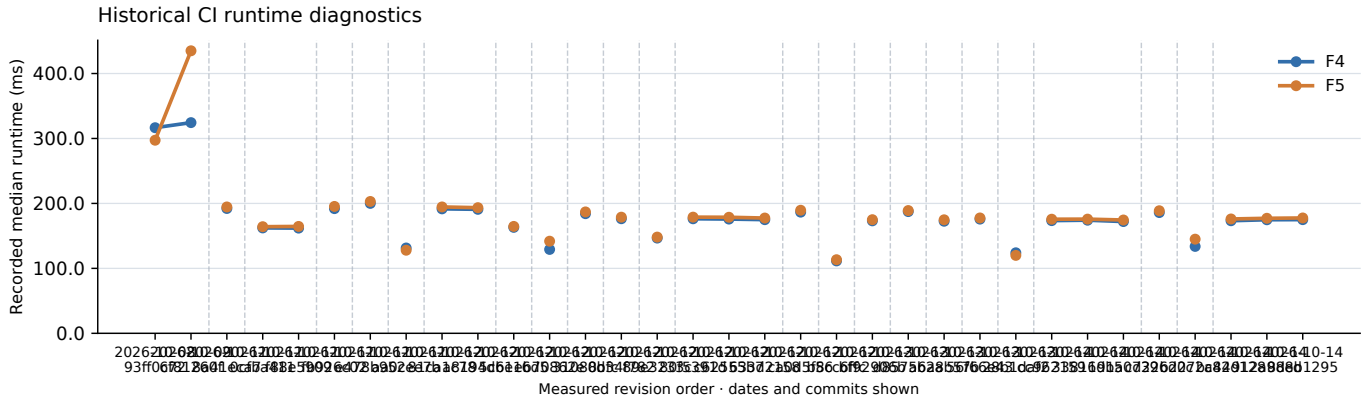
Evidence: 33 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

Boolean solver calls: F4 and F5

Shared K_1 / 2^17 m=2 workload, 16 Boolean variables, four planted and four random targets, maximum degree 3. The solve_boolean_system entry point, including root readback.



33 archived revisions; recorded historical profiler scope, worker coverage incomplete.
Gaps mark changed outputs or measurement conditions. Sources: plot-data.json.



33 archived revisions; recorded medians, no stored spread or isolated timing eligibility.
CI runtime diagnostics; sample counts and condition changes retained in plot-data.json.

UTC date / revision	F4 Ir	F5 Ir	F4 ms	F5 ms
2026-10-08 / 93ff0c72	1,930,989,200	1,953,108,992	316.570	297.297
2026-10-09 / 6f81260f	1,929,374,570	1,952,598,745	324.403	435.004
2026-10-11 / 8a41eca7	1,953,021,023	1,974,789,730	192.203	194.561
2026-10-11 / 0cfba411	1,953,020,766	1,975,267,516	162.266	164.161
2026-10-11 / f88e5909	1,949,981,977	1,973,219,339	161.982	164.584
2026-10-11 / fb926c02	1,953,020,766	1,974,789,781	191.988	195.374
2026-10-11 / e478b952	1,962,376,386	1,985,422,672	200.058	202.996
2026-10-11 / aa0cee7b	1,952,402,641	1,975,976,609	131.376	127.933
2026-10-12 / 81ca1c78	1,962,376,386	1,985,933,163	191.630	194.567
2026-10-12 / a8194d61	1,949,694,281	1,972,772,104	190.899	193.455
2026-10-12 / 5cbe16d5	1,961,124,313	1,981,687,200	163.135	164.619
2026-10-12 / eb70862e	1,961,124,313	1,981,686,883	129.169	141.847
2026-10-12 / 31088bfc	1,783,643,232	1,803,568,510	184.204	186.912
2026-10-12 / 0d34f782	1,787,314,371	1,807,797,314	176.437	178.824
2026-10-12 / 89e3233c	1,786,351,448	1,806,842,750	146.640	148.159
2026-10-12 / 80f5391d	1,787,314,371	1,807,244,283	176.353	178.878
2026-10-12 / c62555b7	1,787,314,371	1,807,244,283	175.899	178.689
2026-10-12 / 633d2158	1,786,234,678	1,806,168,901	174.990	177.494
2026-10-12 / ca0d5f86	1,783,289,117	1,802,994,551	186.580	189.492
2026-10-12 / b5cc6ffc	1,785,257,434	1,804,956,249	111.497	113.333
2026-10-13 / bf92906b	1,785,257,434	1,804,956,249	173.194	174.965
2026-10-13 / d85756aa	1,783,003,746	1,802,715,389	187.409	189.039
2026-10-13 / ab28b57b	1,787,965,641	1,807,670,514	172.579	174.785
2026-10-13 / 56f66e43	1,787,969,850	1,807,674,723	175.850	177.627
2026-10-13 / 28b1daf2	1,787,969,850	1,807,674,723	123.983	119.998
2026-10-14 / cc962159	1,787,965,641	1,808,168,023	173.510	175.690
2026-10-14 / 33816db5	1,790,194,799	1,810,456,555	174.166	175.821
2026-10-14 / 191a0739	1,785,256,826	1,805,459,208	172.076	174.475
2026-10-14 / cd226d0c	1,785,256,826	1,805,459,208	185.930	188.735
2026-10-14 / b2272a44	1,787,965,186	1,808,230,025	133.809	144.964

UTC date / revision	F4 Ir	F5 Ir	F4 ms	F5 ms
2026-10-14 / bc82d128	1,787,965,186	1,808,230,025	173.338	176.053
2026-10-14 / 912a98ed	1,784,912,427	1,805,177,722	174.838	177.121
2026-10-14 / 8d8b1295	1,784,912,427	1,805,177,722	175.021	177.613

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

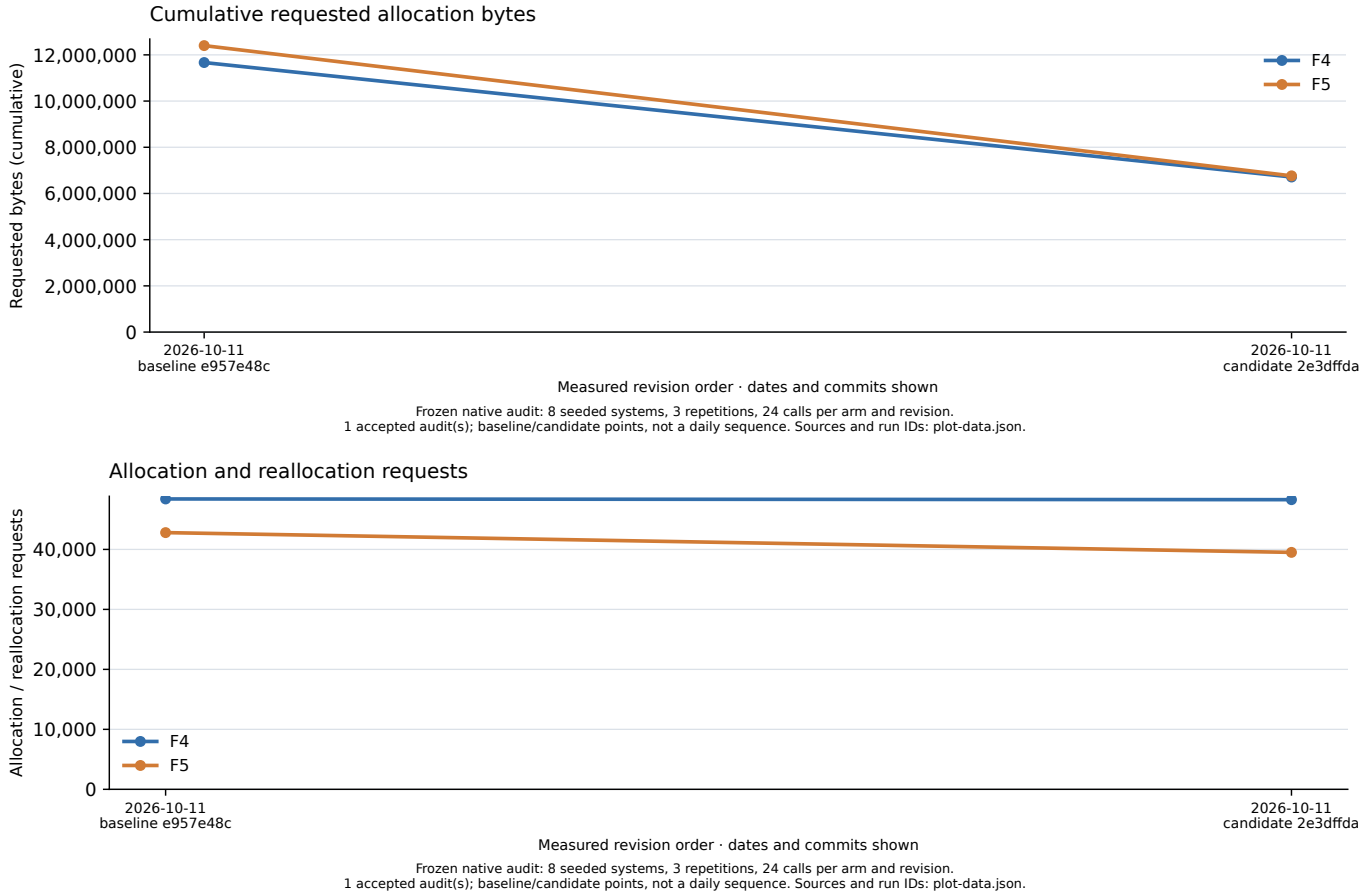
Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

Evidence: 33 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

Complete Boolean calls: allocation history

Matching frozen Boolean allocation suite: 8 seeded systems, 3 repetitions, 24 calls per arm and revision. Polynomial dimensions, degrees and seeds are retained in the source audit protocol.



UTC date / revision	F4 bytes	F5 bytes	F4 requests	F5 requests
2026-10-11 / e957e48c	11,667,696	12,400,920	48,414	42,801
2026-10-11 / 2e3dffda	6,712,608	6,762,696	48,294	39,507

Native allocation audit. Original baseline/candidate rows, accepted comparison, exact outputs and accounting, shared inputs and all totals revalidated by the Rust plot-data tool.

One baseline and one candidate point per accepted audit, not a daily sequence. A/A and replication pairs are controls, not additional algorithm revisions.

Bytes are cumulative requested traffic, not peak memory. No timing is measured by this allocation series.

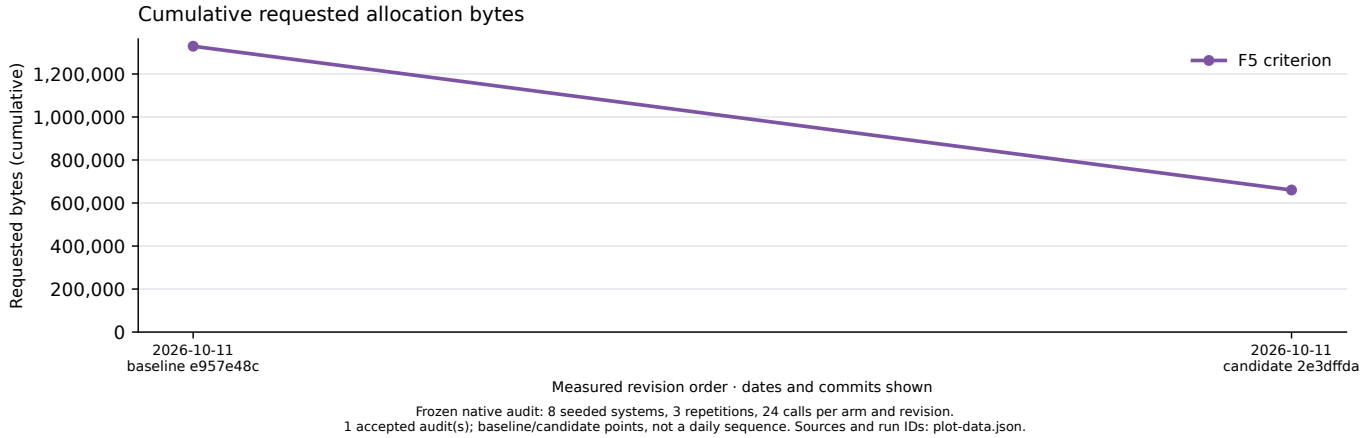
Criterion construction is a separate part of F5 and is never compared as a complete solver.

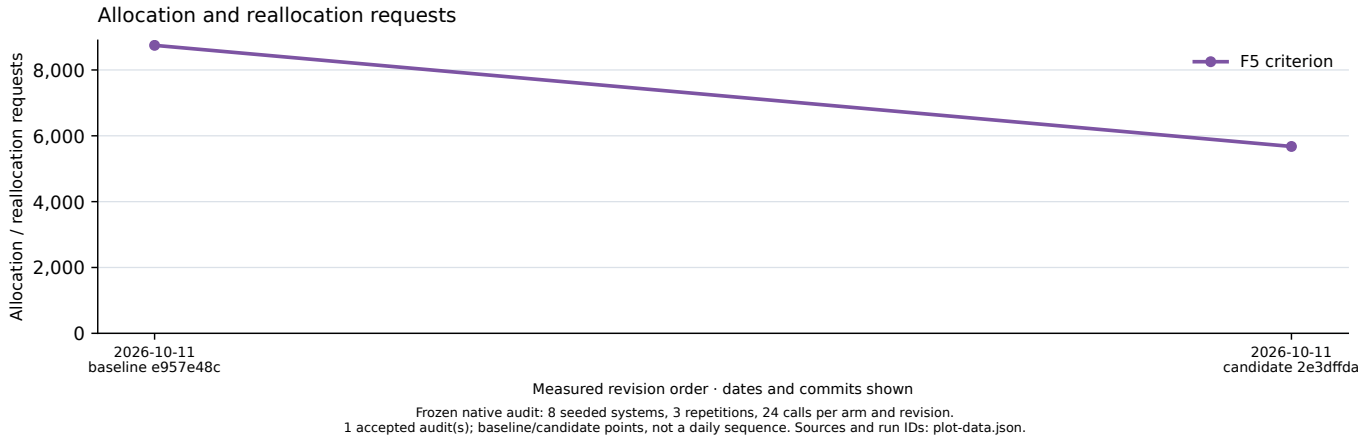
Audit revisions come from their frozen records. They are not inserted as measurements of historical CI workloads; the workloads and metrics remain separate.

Evidence: allocation_sources in plot-data.json; frozen native comparison and baseline/candidate JSONL records in research/f4_f5_row_build_20261011/records/. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

F5 criterion: allocation history

Matching frozen Boolean allocation suite: 8 seeded systems, 3 repetitions, 24 calls per arm and revision. Polynomial dimensions, degrees and seeds are retained in the source audit protocol.





UTC date / revision	F5 criterion bytes	F5 criterion requests
2026-10-11 / e957e48c	1,329,024	8,748
2026-10-11 / 2e3dffda	660,480	5,676

Native allocation audit. Original baseline/candidate rows, accepted comparison, exact outputs and accounting, shared inputs and all totals revalidated by the Rust plot-data tool.

One baseline and one candidate point per accepted audit, not a daily sequence. A/A and replication pairs are controls, not additional algorithm revisions.

Bytes are cumulative requested traffic, not peak memory. No timing is measured by this allocation series.

Criterion construction is a separate part of F5 and is never compared as a complete solver.

Audit revisions come from their frozen records. They are not inserted as measurements of historical CI workloads; the workloads and metrics remain separate.

Evidence: allocation_sources in plot-data.json; frozen native comparison and baseline/candidate JSONL records in research/f4_f5_row_build_20261011/records/. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.